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Pakistan and Cyber Crimes: Problems and Preventions

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Abstract—The growth of internet in Pakistan is exceptional; it reached to the heights of popularity for an ever changing medium of information and communication in a major conservative society. This brings freedom of expression, communication and information across the country which struggled to provide free speech and information access to its people. The internet penetration is increasing very rapidly; it is very important to observe the impact of it on the society; especially when the government is constantly trying to formulate the regulation for controlling the cyber space. A well regulated cyber space leads to the expansion of information technology services speedily. This paper focuses on the prevailing development of governance policies for cyber space and defies among the agencies and information technology vendors. Moreover, the case study of national bank of Pakistan and other cyber crime survey shows that a lot of law making is needed in order to reduce the chances of cyber crime.

Keywords—Cybercrimes, Policies, Trends, Challenges, Preventions

I. INTRODUCTION

The growth of information and communication technologies (ICT) is unprecedented and overwhelming during the past decade. The expansion growth for internet users increased at a rate of 5, 5001.2 percent during 2000 – 2005. The adoption is slow as compared to early rise but still it is increasing at a rate of 4.62 percent, and still it needs to be standardized according to internal standards [1].

This insurrection in information technologies have changed the social order primarily and will possibly go on with it in the near future. Earlier only a few groups of the economic sector had changed its operational procedures due to the advent of ICT, but merely any group of the economic or social society is unaffected with this technology at the moment. It has intensely changed the social structure of today's world [2] [3].

The social and economical changes due these technologies are unparalleled on one hand but as every picture has two sides, the technologies also have a darker side on the other hand as well. It leads to the appearance of new categories of crimes in addition to commit the conventional crimes using emerging technologies. Furthermore, the effects of immoral behavior can be broader as compared to earlier due to

development of ICT which eliminated the concept of geographical boundaries [4].

The governments are encouraging its people to use ICT since its inception, but they are constantly working to counteract the dark side of the technology as well and doing it simultaneously. Legal and technical measures are taken to protect and prevent ICT from catastrophes along with discouraging criminal behavior [4].

Pakistan is also facing the problem of cyber security and is not free from this dilemma. In Pakistan the extraordinary growth, accessibility of computer equipments and growth in internet service providers leads to learn and communicate. On the other hand, a few folks do utilize this powerful technology for unlawful purposes too. In this paper it is tried to explore the problems that are faced and some of the laws that are already available to counter these issues and suggestion which are needed to strengthen these laws.

II. LITERATURE REVIEW

The emerging technologies of mobile and cloud computing, electronic commerce and social applications are primarily changing the world's perspective of business. The business organizations are developing strategies to gain the unfold benefits of this latest technology in information sharing and conducting commerce on the internet [4, 5]. The total volume of business transaction perform online is approximately eighty percent of the total transactions. Therefore, the requirement for high level security is augmented to reduce the chances of transaction failure and customer dissatisfaction. Cyber Security is not limited to securing the local information technology infrastructure of the organization; rather it is extended to much more border network and information technology infrastructure[6]. The role of cyber security cannot be ignored in the current development and deployments of critical computing and communication infrastructure [7]. National security and economical development are the key players for the improvements in cyber security and protection of vital digital infrastructures. It is essential to make the internet more secure and protect the user of internet from cyber attacks in

order to strengthen the services and government policies as well [8].

It is essential to include deterring of cyber crimes as a part of national strategy for protection of computer and communication infrastructure and cyber security. A comprehensive approach is thus needed for the development and implementation of strategies on national scale for cyber security [9]. For instance, the strategies for cyber security including technical protection system implementation or educating the users to protect them against victimization of cyber crime help in the reduction of cyber crimes [10]. These national or individual strategies are essential player in the struggle against the cybercrime. A comprehensive approach is needed to stand against cybercrimes; providing technical solution only, are not enough to avert any crime. The law enforcement agencies should be allowed to effectively and efficiently probe and arraign cybercrime [2, 5, 11].

Cyberspace is the juncture for the criminal groups in the era of organized crime and terrorism; the groups of these kinds are the Russian Mob, Internet Tamil Black Tigers, and Mafia Internet Securities. Cyber terrorism is considered and recognized as cyber crimes. The first terrorist attack of its kind as reported by the intelligence authorities is against computer systems in Sri Lanka. The ethnic guerrillas flooded Sri Lankan embassies emails with over 800 emails in a single day for two week period. The email messages contain the text, "We are the Internet Black Tigers and we are doing this to disrupt your communications". It was a great cause of fear for the diplomats as the rebel group was known for killing people[12].

Anti Cybercrime laws as developed by the international community divided the cybercrimes into nine broad categories. These categories include; privacy protection, harming intellectual property, criminal / procedural laws, internet economic crime, forgery, internet fraud, adolescent pornography, illegal access, illegal / harmful contents.

Despite laws making, a few countries e.g. China and Saudi Arabia have implemented technical procedures of internet filtering. Kingdom of Saudi Arabia established internet service unit (ISU), 2010. ISU is responsible for filtering the web traffic allowing the user to access only the legitimate (the website which are not on the "Black List") web sites [13].

III. EMERGING TRENDS AND CHALLENGES

The main focus of organizations these days is on developing mechanism for protecting the identity of the user and theft of sensitive data. The world is moving towards digital economy and most of the information is in digital form. Social applications of the ICT's in the form of websites for interacting with friends and family where the users feel safe but in reality this is more open to cyber criminals to gain access and steal personal data of the users especially home users. Although

most of the users don't care much about what they are sharing on social networks and the privacy of their post, it can be quite troublesome for them. On top of all benefits of online social networks, there are many disadvantages. Identity theft is one of the biggest concerns of online social networks [14]. There are millions of fake profiles, which maliciously manipulate or harm other people and the reason for this matter is that it is very simple and quite fast to create and form a fake profile by using other's information or picture and use social engineering techniques to steal information.

The devices which work on android operating system will face more attacks, but not on major scale. As it is a known fact that tablets running the same system like smart phones and will also be beleaguered soon by the same kind of malware. Mac would face less malware attacks as compared to window based personal computers. The new window operating system i.e Windows8 facilitates the application developer to virtually develop any application for any devices running Windows8. The possibility of developing malevolent applications is increased as compared to android applications [15, 16]. The above mentioned technologies brought with it; great opportunities and at the same time challenges as well.

The broad distribution of On-line Social Networks, the privacy and confidentiality of the users involved in such services is going to be a key distress. Numerous researchers proposed solutions to lessen the threats on confidentiality and privacy for those group obtaining profiles in online social networks previously (as an instance, count up more than 800 million for Facebook, one of the mainly well-liked online social network). As a case in point of such solutions for privacy-threat mitigation, the concept of Virtual Private Social Network (VPSN) was planned. VPSN fundamentally is a sign of the concept of Virtual Private Network contained by online social networks: only friends surrounded by the VPSN are capable to observe the valid and actual information of an individual [3, 13, 15].

A lot of new apps and devices are developing, which brought with them new security threats. The organization are moving towards paper less economy where everything will be digital. They want to safe guard the sensitive information rather than the physical security of the system only. The companies are trying to get advantages from the cutting edge technology of cloud rather mobile cloud computing but still hesitant to migrate to it because of the security issues relating the privacy and availability [15, 17].

IV. ANTI – CYBERCRIME POLICIES OF THE GOVERNMENT

The cyber security policies of the government should cover the requirements of all citizens (corporate and domestic citizens) not just the government only. To address cybercrimes a legal framework is present in Pakistan. In 2002 Pakistan government has passed the Electronic Transaction Ordinance

(ETO). The main objective of this ordinance is to identify and assist documents, reports, information, and communications, electronic transaction and to provide for accreditation of certification service providers. This legal framework provides legal support for electronic transactions (Information sharing and communication), as a written and signed document. Pakistan is among those countries that have legal framework of this kind which provide essential structure and an impetus for expansion of e – commerce in the country. The ETO is the indispensable prerequisite for electronic commerce expansion and as termed as “a Landmark decision for Information Technology development in the country”. The main aim of this document was to achieve the unprecedented economic impact, promotion of e – commerce while projecting Pakistani products in the global markets, increasing online transaction to greatly reduce the cost of transaction for SME’s, and to protect the buyer and seller interest in e – transactions [18].

The government of Pakistan notified through official gazette, implantation of rules and purpose of the ordinance. The main clauses laid down in the ordinance for offenses related to e – transactions include; provision of false information, false certificate issuance, and damaging information system. In addition to ETO 2002, the ministry of information technology in consultation with the ministry of interiors prepared the Electronic Transaction Act 2004. The ETA (2004) addresses cybercrimes i.e Criminal access, Criminal data access, Data damage, System damage, Electronic fraud, Electronic forgery, Misuse of devices, Misuse of encryption, Malicious code, Cyber stalking, Spamming, Spoofing, Unauthorized interception and Cyber Terrorism [19].

The other steps initiated by Government of Pakistan are the formation of National Response Center and Accreditation Council. The National Response Center is responsible for stopping internet misuse and tracing any involvement in cybercrimes. The accreditation council was formed for Certificate Authorities keeping in view the national policies on information technology ETO 2002. The purpose of licensing program is to promote integrity, honesty, trustworthiness for the licensed certificate authorities [1, 20]. But on the broader picture, these policies are still not mature to counter all the terrorist attack on the cyberspace of the country.

V. DATA COLLECTION

The common methods of data collection used in this research were case studies, questionnaire, emails, phone based and face to face interviews. After collecting the data it was then analyzed using SPSS. The objective of this research was to explore if any or being the victim of cybercrimes and whether they have reported any such incident.

A. On Spot Questionnaires and Emails

The questionnaire was distributed among 3000 users through emails and on spot distribution in different parts of the country i.e majorly in the cities of Peshawar, Lahore, Karachi, Islamabad, Faisalabad, Abbottabad etc. The questions were related to the incidents of abuse / privacy, scams, malware, defacements, hate/ threat mails, unauthorized access, DoS Attacks, Fake Accounts, and Intellectual Property Violations. Out of 3000 users, 1895 have responded and the data were analyzed. The contributors of this study were teachers, college and university students, public servant including police and federal investigation agency officials, researchers and businessmen.

B. Face to Face Interviews

In order to know more about the actual situation of cybercrime records; interview with the official of police department and federal investigation agency (cybercrime wing) were conducted. These interviews are not structured and the purpose of these interviews was to validate the data collected during the survey. The officials pointed out that very less number of the cyber crimes are reported.

VI. RESULT ANALYSIS AND CASE STUDY

It is confirmed through this study that some people using internet in Pakistan are the direct or indirect victims of cybercrimes but most of the cases are not reported to the police. Another case is the case study of National Bank of Pakistan which is a victim of cybercrime itself.

A) Specific Case of Cybercrime

National Bank of Pakistan (NBP) was established in 1949 as the major financial institution of country. The objective of its establishment was to provide all kind of banking services to public and private organization of the country. This bank also serves to cater the treasury transaction on behalf of the govt. as an agent of state bank of Pakistan.

The NBP independently suspended its One-Link service with 14 other banks when it find out that a cyber gang had breached the hardware security modules and PIN codes of the ATM machines in 2008. The NBP called the FIA for legal help and investigation to find out that how the gang is misusing some “zero balance” accounts of the employees among whom, one is already retired.

The official of police arrested Amir Abbas employee of the other bank in Lahore and suspected culprit. This one of the 14 other banks which share the same ATM service. Again in Multan and Lahore the officials of NBP had discovered a “cyber theft” exceeding rupees three (3) millions from its ATMs. This was surprisingly the same incidence as the previous and the NBP management found that similar cases were seen in branches which were operating in the industrial belt of Punjab. During the process of investigation the bank’s

management found that another theft of more than rupees eight (8) millions were detected. This case study shows that Pakistan is facing the great issue of cybercrimes.

B) Results from the Survey

After collected the data it was reported that more than 80 percent of the respondent faced one the cybercrime incident. Most of the victims are complaining about the abuse on social media, fake accounts, virus attacks, defacements and hate / threat emails. All of the internet crimes are not reported as due to non – awareness to the people about the cyber laws in the country. The people are not aware about such kinds of issues and its consequences. The details that were emerged from the user were as follows:

1) The respondent of the survey in response to the use of internet responded that 100 percent of them use it for communication purposes. Another 28 percent of the user's uses internet in Pakistan use the internet for research. 16 percent of the users use internet for business purposes. Thus it is evident that the leading activity on the internet is e-mail and research. The detail of most users' activities on the internet is given in the figure 1.

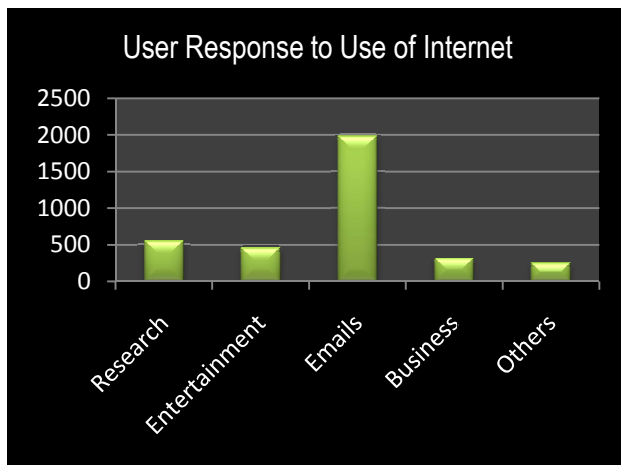


Fig.1. User Response to Use of Internet

2) Majority of the respondent responded that they fall the victim of atleast one cybercrime. The most common cyber crimes investigated in this study are SPAM, defacement, fake accounts, virus attacks, hate / threat emails being the most reported cybercrimes. 58 percent of the victims responded that they never shared it with anybody, 28 percent of respondent reported these to system administrators and only 14 percent of the victims reported it to the law enforcement authorities. Most of the victims would prefer to not tell anyone due to lack of knowledge about the physical evidence and laws relating to these crimes. 35 percent of the responded claimed that they faced hate / threat emails while 23 percent reported the violation of privacy. Defacement and unauthorized access are

reported to be 10 percent of the total. Fake facebook and emails are reported to be 5 percent. These results are depicted in the figure 2.

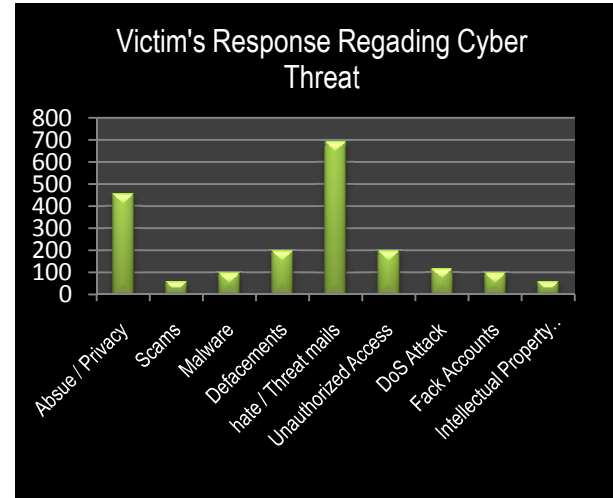


Fig.2. User Response to Cyber Threat

3) In response to the study only 63 percent of the respondent claimed that they are not aware to check any incidence of cybercrimes and most of them don't check their systems for regularity.

The above statistics reveal that the user of internet in Pakistan in one way or the other facing this dilemma of cybercrimes. Even though, public is not reporting these incidence to the concerned authorities due to unawareness about the prevailing cyber laws.

VII. RECOMMENDATIONS AND SUGGESTIONS

After analyzing the results it is believed that prevention is better than cure i.e restricting the number of cyber security violations. Although, it is practically impossible to prevent these incidents completely unless, experience and knowledge in forensic and cyber laws for protecting the victim rights in case of infringements. In this section a few recommendations are presented for the policy makers

A) Sensitization of the Phenomenon of Cybercrimes

This issue should be sensitized. The government should established public awareness program regarding anti – cybercrimes activities. The people need to be trained on the knowledge of internet and computer crimes. The trained user should be able to detect, prevent, minimize and report these incidences. The government should establish; possibly develop facility for the reporting of cybercrimes preferably on the internet. The official of police department should also be trained on investigation process and unified record keeping.

Eforensic is a relatively new field and proper training should be conducted for all the concerned [14, 21].

B) Internet Traffic Filtering

Some countries like China and Saudi Arabia has implemented internet filtering for the clients at the ISPs and government level. The government should prevent illegal access to block website. The websites promoting or having keywords like terrorism, bomb making recipes and pornography should be blocked. Proper legislation is needed for filtering the incoming internet traffic prior to its access by the internet user. The government should block the website which posed to be an eminent threat to the country. The ISPs should be directed to install preventive measures against DDoS, and spamming [13].

C) Enforcement of Cyber Security Laws

Pakistan is among those countries that have passed cyber laws for protecting and promoting of the electronic transaction. The people committing cybercrimes should be dealt with iron hands and the laws should be amended accordingly. This will not only prevent the fast rate of cybercrime but will also motivate the victims to report more about the cybercrimes.

It seems that the laws that are meant to protect the cyber community from intended crimes; are not sufficient to safeguard the non – commercial problems especially the use of internet for spreading of extremist ideologies in Pakistan. The extremists performed unlawful activities on the social media worldwide and Pakistan in particular. These activities not only disturb the autonomy and trustworthiness of individuals but institutions as well. It requires exhaustive policies to cope with cyber terrorism and cyber extremism. These laws could be synchronized with global community in combating terrorism rather cyber terrorism.

Some other recommendations are to extend the development of specialist police and forensic computing resources. Support the international Computer Emergency Response Team (CERT) community, including through funding, as the most likely means by which a large-scale Internet problem can be averted or mitigated. Fund research into such areas as: Strengthened Internet protocols, Risk Analysis, Contingency Planning and Disaster Propagation Analysis, Human Factors in the use of computer systems, Security Economics [10, 22].

VIII. CONCLUSION

It is shown that in one way or the other, cybercrimes are silently present and pose a threat to the developed and developing countries. The instances of cybercrimes are not reported rather discussed socially; on the other hand the perpetrators are invisible under the cyber world. This is a crime

beyond the border so it is hard to bring the cyber criminal under justice system. The laws needed to be modified and cooperation with international community in this regard should be established. It is also hard to obtain facts from internet or computer crime that are needed to prove someone a culprit in a court of justice. Computer forensic is still a new area and mostly the countries are lacking the expertise and literature. It is an eminent threat and the countries like Pakistan should adopt the proper laws and procedures of anti – cybercrimes effectively.

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